

Protocol for QA Test Apparatus

Files: The folder to open in the Arduino IDE with the latest firmware is titled "Urinator4000.ino" and is located in the "internal-utilities" GitHub repository.

How the Test Works: The test first has a cooldown period specified in the "setup_time" variable prior to initiating the first experiment. Once the cooldown timer has ended, the white LEDs will start blinking and after a small delay the motor should activate. The motor and LEDs are activated for a duration of time specified in the "void_duration" variable. Next, the LEDs and motors are deactivated and the system is idle for a duration of time specified in the "cycle_duration" variable. The system then cycles between voiding and idling until the experiment is complete for which the total duration is specified in the "test_duration" variable. Throughout the experiment there are indicator LEDs located on the face of the PCB (red, yellow, green). The meaning of the indicators are provided below:

Indicator LEDs:

Solid red LED = system is in cooldown setup period

Solid green LED = system is in active voiding period

Solid yellow LED = system is idling in between voiding periods

Blinking red LED = system has concluded the test

To Initiate Test: Plug in the 12V power cable first, and then plug in the USB to the Arduino. Ideally, the USB should be plugged into the computer with the Arduino IDE open and the Serial Monitor displayed to show the logs in real time, however, if the USB is connected to a simple power source the device will still run the same. *Note: Ensure that the IDE has Arduino Nano selected as the board and the processor is ATmega328P*

To Modify Test: The parameters for the timings are easily adjustable from the Arduino IDE, just make sure to upload the code to the board before running the test again.

To Restart Test: Either disconnect the USB cable from the Arduino and reconnect it or you can press the reset button on the Arduino (shown below):

